

Full Stack Development with MERN

Project Documentation

1. Introduction

- **Project Title:** DocSpot
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2. Project Overview

The Online Bookstore aims to provide users with a seamless and DocSpot (MedicareBook) is a web-based appointment booking system that connects patients, doctors, and admins in a single digital platform

The system supports:

- Patient appointment booking workflow
- Doctor verification and schedule management.
- Admin supervision of doctors and appointments
- Document upload during booking
- Status updates (Pending → Approved)

Key Features:

Landing Page

- Hero section with “Book Your Doctor” button
- Navigation: Home | Login | Register
- Professional medical UI

Authentication

- Login for Patients
- Login for Doctors
- Login for Admin
- Registration system for new users

3. Architecture

Frontend (React.js)

- Folder structure:

```
client/  
├── components/  
├── admin/  
└── user/
```

- Axios for communicating with backend
- JWT token stored in localStorage
- Role-based UI rendering

Backend (Node.js + Express.js)

Folder structure:

```
server/  
├── controllers/  
├── middlewares/  
├── models/  
├── routes/  
├── uploads/  
├── config/  
└── server.js
```

- Controller-based logic
- JWT authentication
- Multer for file uploads

Database (MongoDB)

Collections:

Users

Doctors

Appointments

4. Setup Instructions

Prerequisites:

- Node.js (LTS version)
- MongoDB Community Server or MongoDB Atlas (Cloud DB)
- Visual Studio Code

Installation

Step 1: Install Dependencies

Frontend

Cd client

Npm install

Backend

Cd ../server

Npm install

5. Folder structure

Frontend

Client/

├── components/ # Shared components (Navbar, DoctorCard, AppointmentCard)

├── admin/ # Admin screens (Doctor verification, Appointment monitor)

├── user/ # Patient + Doctor screens

└── src/

 ├── App.jsx

 └── main.jsx

Backend

Server/

├── config/ # Database connection

 └── connect.js

├── controllers/ # Business logic (adminC, userC, doctorC)

 ├── adminC.js

 ├── doctorC.js

 └── userC.js

└── middlewares/ # Auth and role-based access

```

|   └── authMiddleware.js
|
|── models/      # Mongoose Schemas (Users, Doctors, Appointments)
|
|   └── ...
|
|── routes/      # API route definitions
|
|── uploads/     # Document uploads via multer
|
|── server.js    # Entry point
|
└── .env

```

6. Running the Application

Backend: from server/ run nodemon index.js (or npm start). Frontend: from client/ run npm start. Access the app at <http://localhost:8001>

7. API Documentation

- POST /api/auth/register – register patient or doctor user.
- POST /api/auth/login – login and return JWT + role.
- GET /api/doctors – list approved doctors for patient home page.
- POST /api/doctors/apply – doctor submits “Apply for Doctor” form.
- GET /api/appointments – get appointments for logged-in user (patient or doctor).
- POST /api/appointments – create appointment with selected doctor and date/time.
- PUT /api/appointments/:id/approve – doctor or admin approves appointment.
- GET /api/admin/doctors – admin view of all registered doctors.
- GET /api/admin/appointments – admin view of all appointments.

8. Authentication

- JWT-based login with separate roles: Patient, Doctor, Admin.
- Protected routes on backend and frontend, only allowing access to the correct dashboard based on role.
- Passwords stored as salted hashes; login returns token saved in HTTP-only cookie or local storage.

9. User interface

MediCareBook

🔔 Bruce banner

Apply for Doctor

Personal Details:

* Full Name:

* Phone:

* Email:

* Address:

Professional Details:

* Specialization:

* Experience:

* Fees:

* Timings: →

Submit

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MediCareBook

🔔 Yaksha

Dr. Irshath sher

Phone: 7550088600

Address: 67, 4th st, d

Specialization: psycho

Experience: 3 Yrs

Fees: 3000

Timing: 07:05 : 14:12

Book Now

Booking appointment

Doctor Details:

Name: Bruce banner

Specialization: heart surgeon

Appointment Date and Time:

Documents

No file chosen

Close Book

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MediCareBook

🔔 welcome.. Yaksha

All Appointments for Admin Panel

Appointment ID	User Name	Doctor Name	Date	Status
69330f67fe64e02e382d5e66	Yaksha	Irshath sherine	2025-12-20 12:00	approved
69330f6efe64e02e382d5e6a	Yaksha	Irshath sherine	2025-12-20 12:00	approved
6933922890be51d574656a85	Yaksha	Irshath sherine	2025-12-06 10:50	pending

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10 Testing

- Manual testing using Postman for API endpoints (auth, doctors, appointments, admin).
- UI tested in Chrome and mobile browser, covering flows: register → login → browse doctors → book appointment → doctor approve → admin view.
- Negative tests: invalid login, booking without required fields, booking past date, uploading unsupported document type.

11. Screenshots or Demo :

demo link: [DocSpot](#)

12. Known Issues

- No integration with payment gateway yet; consultation marked as booked without online payment.
- Email/SMS notifications may be simulated or basic instead of production-grade service.
- No advanced search (e.g., fuzzy matching or rating filters). Limited validation on uploaded documents (file type/size).

13. Future Enhancements

- Online payment integration for paid consultations.
- Automated email/SMS reminders using a notification service.
- Doctor ratings and reviews with analytics dashboard.
- Video consultation integration.
- Multi-clinic support and exportable reports for admins.